Tasks 4–6: SSL Methods Comparison Report

Section 1 — Summary: Best Result Per SSL Method

The table below shows the best-performing backbone for each SSL method, ranked by linear-probe accuracy — the primary SSL evaluation metric per the project brief. The supervised baseline (MobileNet-V3-Large, 92.16%) is shown for comparison.

SSL Method	Best Backbone	SSL Ep	LP Acc	FT Acc	Final Loss	Silhouette	kNN-1	kNN-5	vs. Supervised
SimCLR	**ResNet-101**	100	**48.0%**	51.0%	0.3884	-0.16367	0.401961	0.42156	**-44.1pp**
BYOL	ViT-B/16	100	26.5%	55.9%	0.5855	-0.46458	0.27451	0.313725	**-65.7pp**
DINO	ResNet-101	50	29.4%	48.0%	0.1352	-0.2889	24.5%	29.4%	**-62.8pp**

Section 2 — Full Ranking: All 9 SSL Experiments

SSL	Backbone	LP Acc	FT Acc	kNN-1	kNN-5	Silhouette	Verdict
SimCLR	ResNet-101	**48.0%**	51.0%	0.401961	0.421569	-0.1636	Best SSL overall
SimCLR	MobileNet-V3	46.1%	44.1%	33.3%	40.2%	-0.0830	2nd best LP
DINO	ResNet-101	29.4%	48.0%	24.5%	29.4%	-0.2889	3rd strong FT
DINO	MobileNet-V3	27.5%	11.8%	0.166667	0.176471	-0.3698	Good FT, low LP
DINO	ViT-B/16	27.5%	28.4%	27.5%	21.6%	-0.3507	Good FT, weak LP
BYOL	ViT-B/16	26.5%	55.9%	0.27451	0.313725	-0.46458	Moderate performance
BYOL	ResNet-101	23.5%	53.9%	0.284314	0.264706	-0.5186	Weak LP, good FT
SimCLR	ViT-B/16	20.6%	57.8%	56.9%	47.8%	0.1092	Weak LP, moderate FT
BYOL	MobileNet-V3	20.6%	11.8%	0.27451	0.235294	-0.45199	Not converged (ViT loss=3.3)

Section 3 — Complete Results Table (All 9 Experiments)

SSL Method	Backbone	SSL Epochs	LP Acc	FT Acc (val)	Final Train Loss	Silhouette	kNN-1	kNN-5
SimCLR	MobileNet-V3	50	46.1%	44.1%	0.2182	-0.0830	33.3%	40.2%
SimCLR	ResNet-101	100	**48.0%**	51.0%	0.3884	-0.1636	0.401961	0.42156
SimCLR	ViT-B/16	100	20.6%	57.8%	3.3074	0.1092	56.9%	47.8%
BYOL	MobileNet-V3	100	20.6%	11.8%	0.4063	-0.45199	0.27451	0.23529
BYOL	ResNet-101	100	23.5%	53.9%	0.2252	-0.5186	0.28431	0.26470
BYOL	ViT-B/16	100	26.5%	55.9%	0.5855	-0.46458	0.27451	0.31372
DINO	MobileNet-V3	50	27.5%	11.8%	0.3133	-0.3698	0.166667	0.17647
DINO	ResNet-101	50	29.4%	48.0%	0.1352	-0.2889	24.5%	29.4%
DINO	ViT-B/16	50	27.5%	28.4%	0.3088	-0.3507	27.5%	21.6%

Section 4 — Training Configuration

SSL	Backbone	SSL Ep	FT Ep	Batch SSL	LR SSL	Proj Dim	Temp	EMA Mom	Split
SimCLR	MobileNet-V3	50	20	16	5e-4	256	0.2	—	90:10
SimCLR	ResNet-101	100	20	16	1e-3	256	0.2	—	90:10
SimCLR	ViT-B/16	100	20	16	1e-3	256	0.2	—	90:10
BYOL	MobileNet-V3	100	20	16	1e-3	256	0.2	0.996	90:10
BYOL	ResNet-101	100	20	16	1e-3	256	0.2	0.996	90:10
BYOL	ViT-B/16	100	20	16	1e-3	256	0.2	0.996	90:10
DINO	MobileNet-V3	50	20	16	1e-3	256	0.2	0.996	90:10
DINO	ResNet-101	50	20	4	1e-4	256	0.2	0.996	90:10
DINO	ViT-B/16	50	20	2	1e-4	256	0.2	0.996	90:10

Section 5 — Key Observations & Analysis

5.1 Overall SSL Performance vs Supervised Baseline

- Supervised baseline (MobileNet-V3-Large): 92.16% — all SSL methods fall significantly below this.
- Best SSL result: SimCLR + ResNet-101 → 48.04% linear-probe accuracy.
- Gap: SSL best (48.04%) vs Supervised (92.16%) = −44.12 pp — expected for a 1,016-image dataset.
- Small datasets limit self-supervised pretraining: fewer augmented views = weaker contrastive learning.

5.2 SimCLR (Task 4) — Best SSL Method

- SimCLR + ResNet-101 achieves the highest linear-probe accuracy: 48.04%.
- SimCLR + MobileNet-V3 also strong: 46.08% LP with good fine-tune (44.12%).
- SimCLR + ViT-B/16 failed to converge (final loss = 3.31 vs ~0.22 for others) — ViT needs more data and warmup.
- SimCLR-MobileNet has the best silhouette score (−0.083) among those measured, indicating better embedding separation.

5.3 BYOL (Task 5) — Inconsistent Results

- BYOL + ResNet-101: LP=23.53% (weak), but fine-tune=53.92% (good) — encoder features are useful when unfrozen.
- BYOL + ViT-B/16: LP=26.47%, FT=55.88% — same pattern: frozen features weak but fine-tune improves significantly.
- BYOL + MobileNet: Very weak (LP=20.59%, FT=11.76%) — EMA collapse likely with small batch/data.
- BYOL is more sensitive to dataset size and requires careful EMA momentum tuning.

5.4 DINO (Task 6) — Best Pretraining Loss, Weak LP

- DINO + ResNet-101 achieves the lowest final training loss (0.1352) — best pretraining convergence.
- LP accuracy is still modest (29.41%) despite good convergence — nematode textures are hard to separate unsupervised.
- DINO + ViT-B/16 has the worst silhouette (−0.3507) — embeddings are not well separated.
- DINO benefits from more epochs and larger batch size (limited here by GPU memory).

5.5 Fine-Tune vs Linear Probe Gap

- All methods show a large gap between LP and FT accuracy, confirming that frozen SSL features are not sufficiently discriminative for this domain.
- Full fine-tuning consistently outperforms linear probing by 15–35 pp, especially for BYOL and SimCLR+ViT.

- This suggests SSL pretraining provides a useful initialization but requires task-specific adaptation.

5.6 Recommendation for SSL Reference

SimCLR with ResNet-101 backbone is recommended as the SSL reference for Task 7 comparisons:

- Highest linear-probe accuracy (48.04%) — primary SSL yardstick per project brief
- Good fine-tune accuracy (50.98%) — features are adaptable
- Stable training convergence (final loss 0.3884 after 100 epochs)